

1. Create Employee DataFrame

```
employee_data = [  
    (101, "Alice", 25, "HR", 45000,  
    "Chennai", "2022-01-15", 201),  
    (102, "Bob", 30, "IT", 70000,  
    "Bangalore", "2021-06-20", 202),  
    (103, "Charlie", None, "IT", None,  
    "Chennai", "2023-03-12", 202),  
    (104, "David", 28, "Finance", 65000,  
    "Mumbai", "2020-09-18", 203),  
    (105, "Eva", 35, "HR", 80000, None,  
    "2019-05-25", 201),  
    (106, "Frank", 29, "Marketing",  
    55000, "Hyderabad", None, 204),  
    (107, "Grace", 31, "Finance", None,  
    "Pune", "2022-12-01", 203),  
    (108, "Henry", 26, "IT", 60000,  
    "Bangalore", "2024-01-10", None)  
]
```

```
employee_columns = [  
    "emp_id",  
    "name",  
    "age",  
    "department",  
    "salary",  
    "city",  
    "joining_date",  
    "manager_id"  
]
```

```
employee_df =  
spark.createDataFrame(employee_data,  
employee_columns)
```

```
employee_df.show()  
employee_df.printSchema()
```

2. Create Department DataFrame

```
department_data = [  
    ("HR", "Chennai"),
```

```
("IT", "Bangalore"),  
("Finance", "Mumbai"),  
("Marketing", "Hyderabad"),  
("Sales", "Delhi")  
]
```

```
department_columns = [
    "dept_name",
    "location"
]
```

```
department_df =  
spark.createDataFrame(department_data,  
department_columns)
```

```
department_df.show()
department_df.printSchema()
```

3. Create Manager DataFrame

```
manager_data = [
```

```
(201, "Robert"),  
(202, "Jennifer"),  
(203, "Michael"),  
(204, "Sophia")  
]
```

```
manager_columns = [  
    "manager_id",  
    "manager_name"  
]
```

```
manager_df =  
spark.createDataFrame(manager_data,  
manager_columns)
```

```
manager_df.show()  
manager_df.printSchema()
```

4. Additional Employee DataFrame (For Union Questions)

```
new_employee_data = [
```

```
(109, "Irene", 27, "HR", 52000,  
"Chennai", "2024-03-11", 201),  
(110, "Jack", 34, "Sales", 72000,  
"Delhi", "2021-11-20", 205)  
]
```

```
new_employee_df =  
spark.createDataFrame(  
    new_employee_data,  
    employee_columns  
)
```

```
new_employee_df.show()
```

Part A – DataFrame Creation

1. Create the Employee DataFrame using a list of tuples.
2. Create the Department DataFrame.
3. Create the Manager DataFrame.
4. Display the schema of all DataFrames.
5. Display the first 5 rows.

Part B – Select Operations

6. Select only emp_id, name, and salary.
 7. Select all employees from the IT department.
 8. Select employee names and alias the column as Employee_Name.
 9. Select salary after increasing it by 10%.
 10. Select employees whose city is Chennai or Bangalore.
-

Part C – withColumn()

11. Create a new column called bonus equal to 20% of salary.
 12. Create a column called annual_salary.
 13. Create a column called experience_bonus where:
 - Salary > 70000 → 10000
 - Salary between 50000 and 70000 → 5000
 - Otherwise → 2000
 14. Convert employee names to uppercase.
 15. Add a column called country with value "India".
-

Part D – withColumnRenamed()

16. Rename department to dept.
 17. Rename salary to monthly_salary.
 18. Rename multiple columns one by one.
-

Part E – Filter

19. Find employees earning more than 60000.
 20. Find employees older than 30.
 21. Find HR employees in Chennai.
 22. Find employees with NULL salary.
 23. Find employees whose salary is between 50000 and 70000.
 24. Find employees whose names start with "A".
 25. Find employees whose city is not NULL.
-

Part F – Sort

26. Sort employees by salary ascending.
 27. Sort employees by salary descending.
 28. Sort employees by department and salary.
 29. Sort employees by age descending.
 30. Display top 3 highest-paid employees.
-

Part G – FillNA and DropNA

31. Replace NULL salary with 30000.
 32. Replace NULL city with "Unknown".
 33. Replace NULL age with average age.
 34. Replace multiple NULL columns in one statement.
 35. Drop rows containing any NULL values.
 36. Drop rows where salary is NULL.
 37. Drop rows only if all values are NULL.
-

Part H – Date Functions

38. Convert `joining_date` to `DateType`.
 39. Extract joining year.
 40. Extract joining month.
 41. Extract joining day.
 42. Find employees who joined after 2022.
 43. Calculate years worked.
 44. Find employees who joined this year.
 45. Display current date.
 46. Display current timestamp.
 47. Add 30 days to joining date.
 48. Find the difference between joining date and today's date.
-

Part I – Joins

49. Perform an inner join between Employee and Department.
50. Perform a left join.
51. Perform a right join.
52. Perform a full outer join.
53. Join Employee with Manager DataFrame.
54. Display employee name along with manager name.

55. Find employees whose department location matches their city.
56. Find departments having no employees.

Part J – Union

Create another Employee DataFrame:

emp_id	name	age	department	salary	city	joining_date	manager_id
109	Irene	27	HR	52000	Chennai	2024-03-11	201
110	Jack	34	Sales	72000	Delhi	2021-11-20	205

57. Union the two DataFrames.
58. Count total employees after union.
59. Remove duplicate employees after union.
60. Union by column names.

Part K – Other Transformations

61. Display distinct departments.
62. Count employees in each department.
63. Calculate average salary by department.
64. Find maximum salary by department.
65. Find minimum salary by department.
66. Find total salary paid in each department.
67. Count employees in each city.
68. Display unique cities.
69. Remove duplicate records.
70. Drop the manager_id column.

Part L – String Functions

71. Convert employee names to lowercase.
72. Find the length of each employee name.
73. Extract the first three letters of employee names.
74. Concatenate employee name and city.
75. Replace all occurrences of "a" with "@" in names.

Part M – Aggregations

76. Find total salary.
77. Find average salary.
78. Find highest-paid employee.
79. Find lowest-paid employee.
80. Count employees with salary greater than 50000.

Part N – Advanced Questions

81. Display employees whose salary is above the department average.
82. Find the second highest salary.
83. Find duplicate cities.
84. Display employees who joined in the last two years.
85. Replace NULL manager IDs with 999.
86. Create a salary grade:
 - A → Salary $\geq$ 80000
 - B → Salary $\geq$ 60000
 - C → Salary $\geq$ 40000
 - D → Otherwise
87. Create an employee ID string like EMP101.
88. Find employees whose names end with "e".
89. Display employees sorted by joining date.
90. Find employees who have worked for more than three years.

Challenge Questions

91. Rank employees by salary within each department.
92. Find the top 2 highest-paid employees in every department.
93. Find departments with average salary greater than 60000.
94. Find employees who do not have a manager.
95. Display employees with manager names and department locations in a single DataFrame.
96. Identify employees with missing information (NULL values).
97. Replace missing salaries with the department average salary.
98. Calculate monthly tax as 10% of salary.
99. Find employees who joined on weekends.
100. Create a final report containing:
 - Employee Name

- Department
- Manager Name
- Department Location
- Salary
- Bonus
- Annual Salary
- Years Worked
- Salary Grade